PF Lab 2

Task no.1:- Type & save the following programs in visual studio. Run these programs & observe their output.

P:1

```
#include <iostream>
using namespace std;

int main ( )
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

Output:-
A
B

P:2

```
#include <iostream>
using namespace std;

int main ( )
{
    Char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

output:-
A
B

P:3

```
#include <iostream>
using namespace std;

int main ( )
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

Output:-
65
66

Task no.2:- Take two integers from user apply arithmetic operations on them (+,-,*,/) as print on screen.

```
#include <iostream>
using namespace std;

int main ( )
{
        int a, b;
        cout << "Enter first integer: ";
```

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```
cin >> a;
cout << "Enter second integer: ";
cin >> b;

cout << a << "+" << b << "=" << a+b << endl;
cout << a << "-" << b << "=" << a-b << endl;
cout << a << "*" << b << "=" << a*b << endl;

if (b!=0)
    cout << a << "/" << b << "=" << (double)a/b << endl;
else
    cout << "Division by zero is not allowed." << endl;

return 0;

}
```

Output: (inputs 10 & 4):

```
Enter first integer: 10
Enter second integer: 4
10+4 = 14
10-4 = 6
10*4 = 40
10/4 = 2.5
```

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Task no.3:- Write the output of the following code

```
#= include "stdafx.h"
# include <iostream>
using namespace std;

int main ()
{
    int val1;
    int val2;
    int val3;
    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    system ("PAUSE");
    return 0;
}
```

Output:

```
Please enter your 3 numbers: 6 7 8
6
7
8
8
7
6
```

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Task no.4:- Write a C++ program to calculate the distance b/w two points.

Formula: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

```
#include <iostream>
#include <cmath>
using namespace std;
int main() {
  double x1, y1, x2, y2, distance;
  cout << "Enter x1 and y1: ";
  cin >> x1 >> y1;
  cout << "Enter x2 and y2: ";
  cin >> x2 >> y2;
  distance = sqrt(pow(x2-x1,2) + pow(y2-y1,2));
  cout <<"distance = " << distance << endl;
  return 0;
}
```

Output :

```
Enter x1 and y1: 1 2
Enter x2 and y2: 4 6
Distance = 5
```

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main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

```
A
B
```

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main.cpp

Run ▶

Output

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

```
A
B
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;

int main()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;

    return 0;
}
```

Output

```
65
66
```

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main.cpp

Run ▶

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a,b;
    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << a << " + " << b << " = " << a + b << endl;
    cout << a << " - " << b << " = " << a - b << endl;
    cout << a << " * " << b << " = " << a * b << endl;
    cout << a << " / " << b << " = " << a / b << endl;
    return 0;
}
```

Output

```
Enter two integers: 10 2
10 + 2 = 12
10 - 2 = 8
10 * 2 = 20
10 / 2 = 5
```

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main.cpp

Run ▶

```
#include<iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;
    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    return 0;
}
```

Output

```
Please enter your 3 numbers:2 5 7
2
5
7
7
5
2
```

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main.cpp

Run ▶

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2, distance;

    cout << "Enter x1 and y1: ";
    cin >> x1 >> y1;
    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;

    distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

Output

```
Enter x1 and y1: 1 2
Enter x2 and y2: 4 6
Distance = 5
```

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